

Cyber-Crime and Privacy Violations

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Part 1: Cyber-Crime Incident

International Cyber-Crime Incident

The WannaCry [1] ransomware assault, which occurred in May 2017, is among the most well-known instances of cybercrime. This assault encrypted data on Microsoft Windows machines and demanded Bitcoin ransom payments. The ransomware used EternalBlue, a tool purportedly created by the US National Security Agency (NSA), to take advantage of a flaw in the Windows operating system.

Timeline of Events:

- In March 2017, Microsoft released a fix to address the EternalBlue vulnerability.
- May 12, 2017: More than 230,000 systems in 150 countries were affected within a day of the WannaCry [1] ransomware outbreak beginning.
- Impact: By disrupting government systems, financial institutions, and healthcare services—most notably the National Health Service (NHS) in the United Kingdom—the attack resulted in significant operational and financial losses.

Legal Framework:

International cooperation was required in the WannaCry [1] response. The basis for putting cybercriminals on trial has laid down by the Computer Fraud and Abuse Act (CFAA) in United States and similar laws in other countries. Europol and Interpol were also trying to trace the origin of the attack.

Analysis of Legal Response:

Many organizations failed to upgrade their system even when the patch was released before the attack and this shows that many firms lack adequate knowledge in cybersecurity. Even though legal solutions allowed identifying the culprits, the failure in following the most basic cybersecurity principles prevented the issue from being avoided.

Local Cyber-Crime Incident in Pakistan

Something similar took place in Pakistan even in 2021 where a local banking institution was targeted by ransomware that forced the organization to close its internet for some time. They

collected sensitive information related to clients and encrypted them before asking for money in the form of cryptocurrency.

Legal Framework in Pakistan:

The case was managed under the Prevention of Electronic Crimes Act (PECA) 2016 [2] along with Section 3 – unauthorized access to an information system and Section 6 – malicious code. This case was investigated by the FIA Cyber Crime Wing.

Comparison:

Though people coming from all over the world joined the fight against WannaCry [1], the Pakistani fight against it involved domestic institutions. However, what both cases showed was the importance of having preventive measures in place for example upgrading often and having good IT security policy.

Part 2: Privacy Violation Incident

International Privacy Violation Incident

In 2018, there was the well-known Facebook-Cambridge Analytica Scandal [3] and the company was accused of buying data belonging to tens of millions of Facebook users in complete violation of the law. It was reported that this information was employed for political advertisement, for instance the 2016 US Presidential campaigns and the Brexit process.

Timeline of Events:

- 2014: It was through a third-party app that Cambridge Analytica started collecting data.
- 2018: The scandal was revealed by data scientist Christopher Wylie which caused indignation throughout the world.

Legal Framework:

It compels the Federal Trade Commission (FTC) in the US and legal actions under the General Data Protection Regulation (GDPR [4]) in Europe.

Analysis of Legal Response:

Facebook was penalized in large part because of the GDPR's [4] emphasis on user permission. The event did, however, highlight the absence of international data protection regulations.

Local Privacy Violation Incident in Pakistan

Sensitive information from the NADRA database was purportedly published online in 2020, endangering the personal information of citizens in a similar privacy breach.

Legal Framework in Pakistan:

There is currently no comprehensive data protection law in Pakistan. However, just a portion of the PECA 2016's [2] Sections 41 and 42—which address unauthorized access and misuse of sensitive data—were used.

Suggestions for Improvement:

- **Create Data Protection Laws:** It is necessary to pass a comprehensive regulation like to the GDPR [4].
- **Strengthen Regulatory Oversight:** It is necessary to establish an impartial data protection body.
- **Awareness Campaigns:** Inform the public about their right to privacy and safe internet conduct.

Comparison:

Unlike the GDPR's [4] robust framework, Pakistan's legal system lacks specific legislation to adequately address pervasive privacy violations. This legal loophole makes it difficult to hold perpetrators accountable.

Conclusion

Strong legal frameworks and public awareness are necessary to address global challenges like cybercrime and privacy violations. Although countries such as the United States and the European Union have established mechanisms to address such issues, Pakistan must strengthen its legal framework to effectively protect its citizens. By learning from global events like WannaCry [1] and Cambridge Analytica, Pakistan should improve its cybersecurity and privacy regulations and make the internet a safer place for everyone.

References

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